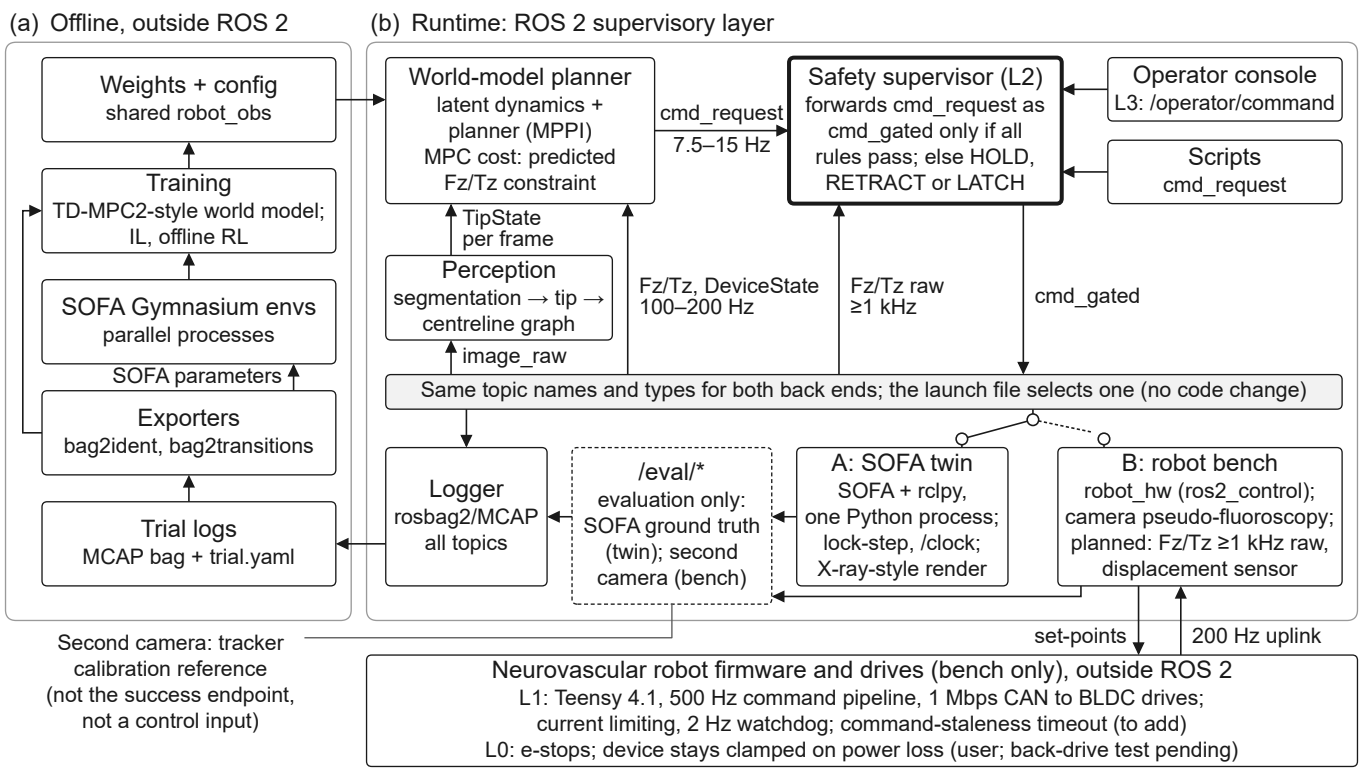
(a) Offline, outside ROS 2
(b) Runtime: ROS 2 supervisory layer
Weights + config
shared robot_obs
Training
TD-MPC2-style world model;
IL, offline RL
SOFA Gymnasium envs
parallel processes
SOFA parameters
Exporters
bag2ident, bag2transitions
Trial logs
MCAP bag + trial.yaml
World-model planner
latent dynamics +
planner (MPPI)
MPC cost: predicted
Fz/Tz constraint
cmd_request
7.5–15 Hz
Safety supervisor (L2)
forwards cmd_request as
cmd_gated only if all
rules pass; else HOLD,
RETRACT or LATCH
Operator console
L3: /operator/command
Scripts
cmd_request
TipState
per frame
Perception
segmentation → tip →
centreline graph
image_raw
Fz/Tz, DeviceState
100–200 Hz
Fz/Tz raw
≥1 kHz
cmd_gated
Same topic names and types for both back ends; the launch file selects one (no code change)
Logger
rosbag2/MCAP
all topics
/eval/*
evaluation only:
SOFA ground truth
(twin); second
camera (bench)
A: SOFA twin
SOFA + rclpy,
one Python process;
lock-step, /clock;
X-ray-style render
B: robot bench
robot_hw (ros2_control);
camera pseudo-fluoroscopy;
planned: Fz/Tz ≥1 kHz raw,
displacement sensor
Second camera: tracker
calibration reference
(not the success endpoint,
not a control input)
set-points
200 Hz uplink
Neurovascular robot firmware and drives (bench only), outside ROS 2
L1: Teensy 4.1, 500 Hz command pipeline, 1 Mbps CAN to BLDC drives;
current limiting, 2 Hz watchdog; command-staleness timeout (to add)
L0: e-stops; device stays clamped on power loss (user; back-drive test pending)